Fixed Income: Credit Risk

LECTURE 11 · REVIEW NOTES
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A bond specifies promised payments. Credit risk is the possibility that the borrower does not deliver those payments in full and on time. I want you to separate three things: the coupon written in the contract, the yield investors require today, and the return investors eventually realize.

1. Start with price, coupon and YTM

$$P_0 = \sum_{j=1}^{2T} \frac{C}{(1+y/2)^j} + \frac{FV}{(1+y/2)^{2T}}, \qquad C = \frac{c \times FV}{2}$$

For a standard fixed-rate bond on a coupon date, a higher required YTM reduces the present value of the same promised cash flows. If YTM exceeds the coupon rate, price is below face value. Equal rates imply par; YTM below coupon implies a premium. Credit news can change YTM without changing the coupon.

2. Measure a credit spread using yields

$$s = y_{corporate} - y_{Treasury}, \qquad 1\% = 100\ bp$$

Compare yields with the same currency, pricing date, similar maturity and yield convention. Do not subtract coupons or bond prices. In the Apple 2035 issue example, the corporate YTM is 4.834% and the Treasury benchmark is 4.334%.

$$s = 4.834\% - 4.334\% = 0.500\% = 50\ bp$$

The Microsoft/Icahn comparison illustrates how borrowers with similar maturities can face very different required yields. Those slide quotes are indicative and not synchronized. A tiny negative measured spread should not be interpreted as negative default risk.

3. Why mortgages and credit cards differ

The slide comparisons give 7.32% - 4.96% = 2.36 percentage points (236 bp) for mortgages, and 22.36% - 4.96% = 17.40 points (1,740 bp) for credit cards. These are broad lending-rate differences. Collateral, prepayment, servicing costs, lender margins and different dates or maturities matter; the entire difference is not pure default risk.

Read the credit information carefully

4. Ratings, default and delinquency

One notch means one step on a rating scale. BBB- is the lowest S&P;/Fitch investment-grade rating; BB+ is speculative grade. Moody's corresponding boundary is Baa3/Ba1. CCC signals severe vulnerability; D indicates default under S&P;/Fitch terminology. An outlook is separate from the rating. Ratings are opinions, not guarantees.

The New York Fed graph measures the share of balances 90+ days delinquent. That stock of overdue balances is not the probability that a borrower defaults over the next year. Similarly, a historical industry default frequency is a starting point, not automatically the PD of the company we are valuing.

5. What goes into a spread?

$$s \approx PD(1 - R) + \text{risk premium} + \text{liquidity premium}$$

PD measures default likelihood over the stated horizon. Recovery R is the fraction recovered if default occurs, so 1 - R is the loss fraction. The other premiums compensate investors for uncertainty and difficulty selling. In our calculations, we set these other premiums to zero and use a constant annual spread.

$$s \approx PD(1 - R), \qquad y = y_T + s$$

Use decimals in the formula: 2% is 0.02. This is an annual teaching approximation, not an exact multi-year pricing identity. A three-year maturity does not mean multiplying the annual spread by three before adding it to annual YTM.

6. High YTM is not a promised realized return

XCCC holds very risky corporate bonds. Its October 27, 2026 snapshot in the lecture shows YTM 17.07%, average coupon 7.76% and OAS 1,196 bp. These describe different quantities. Diversification across issuers reduces concentration but cannot eliminate economy-wide defaults. The fund's past total returns and its current YTM also refer to different periods.

Spirit makes the distinction concrete. Its high pre-bankruptcy promised yield assumed the original payments would arrive in full. After the November 2024 filing, the March 2025 package included about $63 of new debt principal per $100 old face, plus equity. This was not $63 in cash or a measured 63% market-value recovery. The value depends on the new securities. Do not calculate a post-default YTM from a canceled original payment schedule.

From the spread to a bond and a portfolio

7. Work through the issuance example

Given: FV = $1,000; annual coupon = 5%; maturity = 3 years; semiannual payments; Treasury YTM = 4%; annual PD = 2%; recovery = 40%. Other spread premiums are zero.

First, calculate the spread. If default occurs, the loss is 60%. Multiply that loss fraction by the 2% annual probability.

$$s = 0.02(1 - 0.40) = 0.012 = 1.20\% = 120\ bp$$

Next, determine the required YTM. Add the spread to the Treasury yield, using consistent units.

$$y = 4.00\% + 1.20\% = 5.20\%$$

Then identify the cash flows. The annual coupon is $50, so each half-year pays $25. Three years gives six payments. The last payment is $1,025.

$$P_0 = \sum_{j=1}^{6} \frac{25}{1.026^j} + \frac{1{,}000}{1.026^6} = \$994.51$$

In Excel: `=-PV(5.2%/2,3*2,25,1000)`. Price is below par because the required yield exceeds the coupon. Issuance does not force the price to equal face value.

8. Apply the same method to five bonds

For each bond, compute its spread from its own PD and recovery, then add the matching Treasury yield. With 20% of market value invested in each bond, the portfolio measures are simple averages.

$$\bar{s} = \sum_{i=1}^{5} 0.20 s_i, \qquad \bar{y} = \sum_{i=1}^{5} 0.20 y_i$$

For example, spreads of 40, 60, 90, 120 and 140 bp average 90 bp. With a common 4% Treasury benchmark, the average YTM is 4.90%. These are market-value weights, not face-value weights. The weighted-average YTM is not the exact IRR of the portfolio's combined cash flows.

What I want you to retain: measure the spread correctly; keep PD and recovery distinct; translate the spread into a required yield; discount promised payments consistently; and distinguish that quoted yield from the realized return.

Sources used in the lecture: Apple May 2025 pricing terms; New York Fed Household Debt and Credit, 2026 Q2; rating-agency definitions; BondBloxx XCCC portfolio disclosure; Spirit restructuring disclosures. Historical quotes and case figures retain the dates shown in the slides.